

Udayan Atreya

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SUMMARY

AI/ML engineer with hands-on experience building **multimodal AI systems** (audio, video, text), **AI agent frameworks** (reasoning, planning, tool use, multi-agent collaboration), and **production-grade LLM pipelines** on cloud infrastructure. Experienced in LangChain, RAG, LLMs, and agentic architectures; proficient in Python with a strong foundation in deep learning. Pursuing MS in Computer Science (SJSU, May 2026) with coursework in RAG and Applied Deep Learning. Seeking to contribute to AIGC and AI agent systems at NewsBreak's content intelligence platform.

TECHNICAL SKILLS

AI/ML & AIGC: Multimodal AI (audio, video, image, text), LLMs, Diffusion Models (foundational), AI Agents, RAG, AIGC Pipelines, Reasoning & Planning, Multi-Agent Collaboration, Tool Use

Frameworks & Libraries: LangChain, HuggingFace, TensorFlow, PyTorch, RAGAS, FastAPI, .NET

LLMs & AI Tools: GPT-4o, Gemini, Claude, Llama-3, DeepSeek, Ollama, Groq, Whisper (ASR), Vapi, Eleven Labs

Cloud & Infrastructure: AWS (Lambda, S3, API Gateway, DynamoDB, EventBridge), CI/CD, MongoDB, Pinecone, ChromaDB, Prometheus, Grafana

Languages: Python, Java, C#, SQL

Frontend/Integration (exposure): REST API integration, JavaScript, data visualization dashboards

PROJECT EXPERIENCE

Multimodal Evaluator-Optimizer RAG for Language-Based Video Reasoning

[\[GitHub\]](#)

(LangChain, GPT-4o, Gemini 1.5 Pro, Llama-3, ChromaDB, Whisper, PyTorch)

Aug 2025 – Mar 2026

- Architected an **agentic multimodal RAG framework** integrating audio (Whisper ASR), video, and text modalities with an Evaluator-Optimizer feedback loop, **reducing hallucination by 25%** and **improving answer accuracy by 18%** over baselines for Language-Based Video Reasoning (LBVR) tasks.
- Implemented **multi-agent collaboration** between an Evaluator agent and Optimizer agent for iterative response refinement, demonstrating planning and tool-use patterns applicable to production AI agent systems.
- Conducted an **ablation study** on multimodal embedding architectures, showing lightweight models (MiniLM) with agentic feedback achieve **98% performance parity** with SOTA (BGE-base) while **reducing storage costs by 50%**.
- Engineered an end-to-end **RAG evaluation and monitoring suite** using RAGAS and Llama-3-70b (LLM-as-judge) over 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Groundedness, Context Precision/Recall, Noise Sensitivity, and Semantic Similarity.
- Optimized latency-accuracy trade-off with Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while maintaining **90%+ semantic coverage** – a production-readiness consideration for scalable AIGC pipelines.

Agentic Observability Platform with Reasoning and Tool-Use AI Agents

[\[GitHub\]](#)

(LangChain, Gemini 1.5, FastAPI, MongoDB, AWS S3, Prometheus, Grafana)

Jun 2025 – Jul 2025

- Deployed a **production-grade agentic AI system** with semantic routing, **reasoning and planning** capabilities, and real-time observability (Prometheus/Grafana dashboards), achieving **95% query classification accuracy** and **reducing manual triage time by 30%**.
- Designed **tool-use and multi-agent collaboration** patterns: agents autonomously select retrieval tools, escalate to human-in-the-loop validation, and coordinate hardware interventions – directly analogous to AI agent architectures for AIGC content systems.
- Built a hybrid retrieval engine (MongoDB hot store + S3 archive) for scalable data synthesis, **cutting retrieval latency by 75%** (120s to 30s), demonstrating production ML engineering with latency, scaling, and reliability considerations.
- Enforced strict policy checks and role-based access controls on agentic tool invocations, **preventing 100% of unauthorized state changes** – ensuring safe and reliable AI agent deployment.

Analytics-Driven Recommendation Engine (NLP + Personalization)

(Python, AWS Lambda, DynamoDB, NLP scoring, REST APIs)

Nov 2020 – Nov 2021

- Developed a **content recommendation engine** for 100+ automation assets using weighted NLP scoring and personalization logic, **reducing user discovery time by 30%** – a directly applicable foundation for content intelligence systems.
- Integrated recommendation outputs into a user-facing product via REST APIs, exposing AI-driven suggestions to end users and demonstrating **AI-to-product integration** experience.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a **ChatOps AI platform** translating natural language queries into automation workflows via serverless microservices, integrating NLP-driven intent routing – analogous to **AI agent planning and tool-use** pipelines; **reduced operational overhead by 40%** (~5m to ~1m per task).
- Built a user notification system with **production microservices on AWS (API Gateway, Lambda, DynamoDB)** achieving a **20% increase in DAU** post-launch, demonstrating end-to-end AI system deployment with measurable product impact.
- Optimized **DynamoDB access patterns** (schema redesign, pk/sk tuning) and CRUD microservices, **reducing retrieval latency by 37%** and ensuring high-availability workloads – key production-grade ML engineering skills.
- Delivered a **code vulnerability remediation pipeline** using serverless architecture (AWS Lambda, S3, EventBridge, Azure DevOps), safeguarding **400+ applications** – experience with scalable, production-grade AI-adjacent automation systems.
- Designed a single-touch server patching solution using serverless architecture integrated with ServiceNow alerts, **reducing MTTR by 80%**, demonstrating reliable automated decision-making in production environments.
- Collaborated with **cross-functional product, research, and engineering teams** to define requirements and deliver AI-integrated solutions, aligned with the collaborative model at NewsBreak.

EDUCATION

Master of Science, Computer Science

Expected May 2026

San Jose State University, San Jose, CA

Coursework: Retrieval Augmented Generation (RAG), Applied Deep Learning (multimodal architectures, generative models), Biometric Security with AI.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Relational Database Management Systems, Object-Oriented Programming.